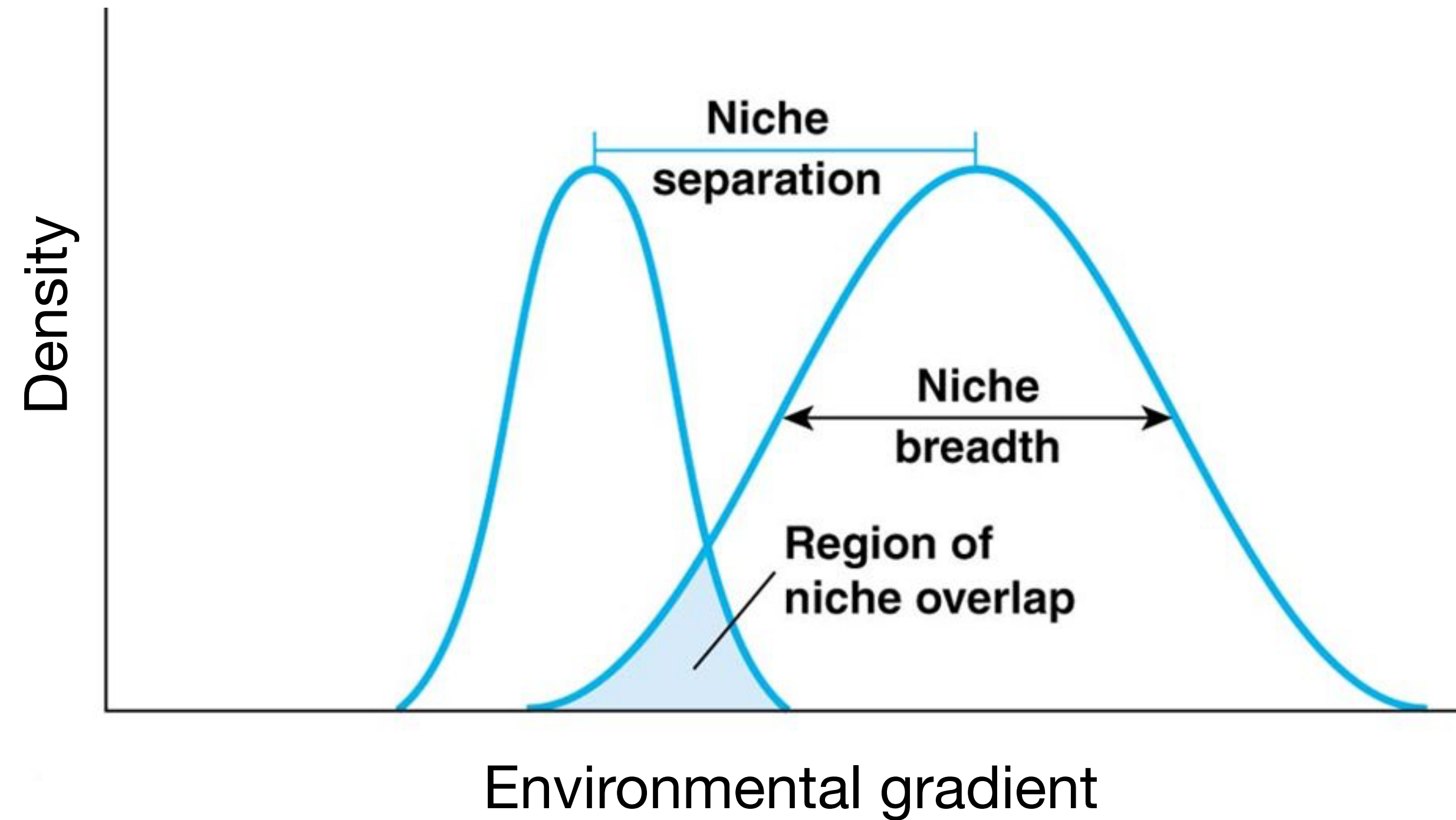


Niche overlap estimates

Modeling the distribution of marine biodiversity under global climate change

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Niche overlap

Niche overlap: the degree to which two species use the same environmental conditions or resources - how much their ecological requirements coincide.

Allow researchers to quantify and compare environmental niches.

Why addressing niche overlap?

Understanding species coexistence and competition: high niche overlap between species can suggest strong interspecific competition.

Investigating speciation and niche evolution: comparing niches can provide insights into speciation processes.

e.g., sister species: did species diverge into different environmental niches (ecological speciation), or do they retain similar niches (niche conservatism), and speciated via other mechanisms (e.g., reproductive isolation)?

e.g., invasive species: comparing the niche of an invasive species in its native versus invaded range can reveal whether the species has shifted its niche to successfully colonize the new environment (niche conservatism).

(...)

Approaches to measuring niche overlap

Geographic overlap (e.g., distribution map comparison);

Physiological responses overlap (e.g., Laboratory experiments);

Environmental overlap (environmental space).

Environmental overlap

Quantitative tools to measure **the degree at which the niche of two species overlap** - based on their environmental distributions.

Schoener's D: focuses on the amount of shared niche space, but giving more weight to high-probability areas - places where species are most likely to be.

Hellinger's I: transforms probabilities using a square root, reducing the influence of extreme values (like very high suitability) - **softer measure** that gives a constant weight across the two species environmental space.

Both D and I provide a value **between 0 (no overlap) and 1 (complete overlap)**, making it easy to interpret and compare.

Environmental overlap

Equivalency test: are the species using the same niche?

Similarity test: are the species' niches more similar than expected by chance?

If $p < 0.05$, the difference (or similarity) is unlikely due to chance (i.e., different niches).

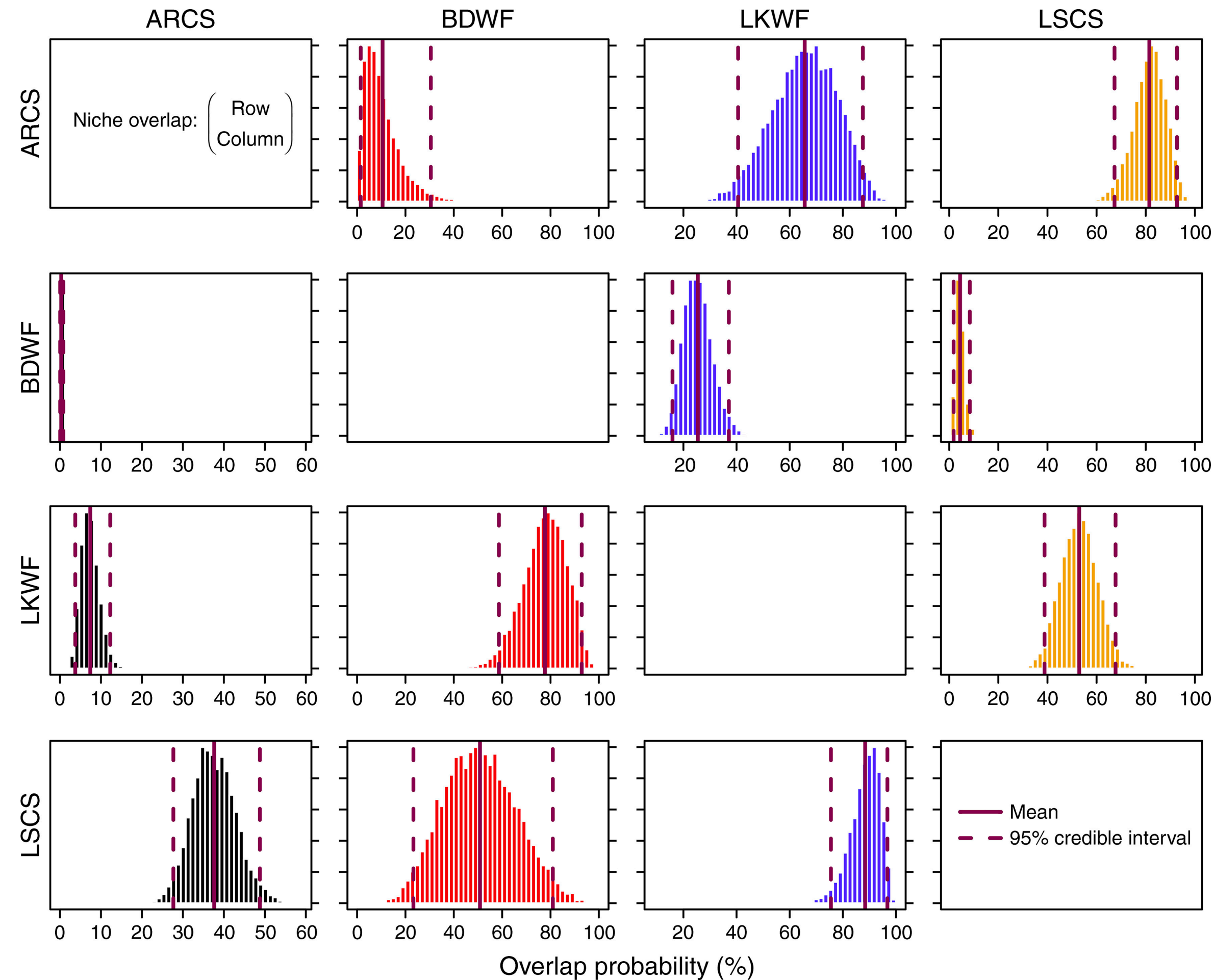
If $p > 0.05$, the observed overlap is within the range of random expectation.

Probabilistic method for quantifying n-dimensional niches

Defines overlap as the probability that individuals from species A occur in B's region.

Directional, multivariate, Bayesian framework.

Consistent with Hutchinson's n-dimensional hypervolume.



Distribution of the probabilistic niche overlap metric (%) for a specified niche region - **probability of species displayed in rows overlapping onto those displayed in columns**. Mean and 95% intervals of niche overlap in purple.

References

Swanson et al., 2015: <https://doi.org/10.1890/14-0235.1>

Schoener, 1968: <https://doi.org/10.2307/1935534>

Warren et al., 2018: <https://doi.org/10.1111/j.1558-5646.2008.00482.x>